Nhat Minh Doan (Đoàn Nhật Minh)

Affiliations Institute of Mathematics, Vietnam Academy of Science and Technology (IMVAST)
Email: dnminh@math.ac.vn
Homepage: https://sites.google.com/view/dnminh

Research Area Hyperbolic Geometry, Low-Dimension Topology. Currently, I am developing a program on the multiplicity of simple closed geodesics on hyperbolic surfaces.

Education

PhD in Mathematics, University of Luxembourg, Luxembourg 2018 - 2022
Advisor: Prof. Hugo Parlier
Thesis: Quantifying some properties of curves and arcs on hyperbolic surfaces

MSc in Mathematics, University of Nice Sophia Antipolis, France 2016 - 2017
Advisor: Prof. François Labourie
Thesis: Arithmetic and ergodic theory of hyperbolic surfaces

BA in Mathematics, University of Education, Ho Chi Minh City, Vietnam 2013 - 2015

Academic Positions

[3] **Researcher**, IMVAST, Vietnam 2022 - now

[2] **SASEA Research Fellow**, NUS, Singapore 2024 - 2026
Advisor: Prof. Ser Peow Tan

[1] **Doctoral Researcher**, University of Luxembourg, Luxembourg 2018 - 2022

Papers

[10] McShane-Rivin norm balls and simple-length multiplicities 2026
(with Xiaobin Li, and Van Nguyen)
arXiv:2605.14574

[9] Graded Bridgeman dilogarithm identities on hyperbolic surfaces 2026
(with Ara Basmajian, Hugo Parlier, and Ser Peow Tan)
arXiv:2601.03876

[8] Optimal special polygon for the congruence subgroup $\Gamma_0(2^n)$ 2026
(with Sang-hyun Kim, Mong Lung Lang, Ser Peow Tan)
arXiv:2601.01324

[7] Some arithmetic aspects of ortho-integral surfaces 2025
(with Khanh Le)
Trans. Am. Math. Soc., https://doi.org/10.1090/tran/9657

[6] Some remarks on subgroups of free-by-cyclic groups 2024
(with Hoang Thanh Nguyen)
Vietnam J. Math., 54, 715 - 728 (2026)

[5] Self-intersection of arcs on a pair of pants 2024
(with Hanh Vo)
BLMS, 57 (2025), 2190 - 2209

[4] Optimal special polygons for the congruence subgroups $\Gamma_0(p)$ and $\Gamma_0(pq)$ 2022
(with Sang-hyun Kim, Mong Lung Lang, and Ser Peow Tan)
J. Geom. Anal., Volume 35, 216 (2025)

[3] Ortho-integral surfaces 2021
Adv. Math., Volume 466, 2025, 11016

[2] Geometric filling curves on punctured surfaces 2020
Glasg. Math. J., Volume 65, Issue 2, May 2023, pp. 383 - 400

[1] Measuring Pants 2020
(with Hugo Parlier and Ser Peow Tan)
Trans. Am. Math. Soc. 376 (2023), 5281-5306

Referee work IMRN, Proc. AMS, Math. Ann., Trans. AMS, Conform. Geom. Dyn.,...

Talks

[27] Seminar, TIMAS, Thang Long University, Vietnam	June 2026
[26] Topology and Geometry Seminar, NUS, Singapore	May 2026
[25] Topology and Geometry Seminar, Southwest Jiaotong University, China	March 2026
[24] Workshop on Geometry and related topics, Phu Tho, Vietnam	Jul 2025
[23] Grad Student Seminar, NUS, Singapore	Apr 2025
[22] Teichmüller Theory and Flat Structures, Matrix, Creswick, Australia	Mar 2025
[21] Workshop on Dynamics and Related Topics, IMS, NUS, Singapore	Mar 2025
[20] Workshop on Geometry and Dynamics in Low Dimensions, IMVAST, Hanoi	Jan 2025
[19] Scientific workshop for young researchers, Vinh Phuc, Vietnam	Sep 2024
[18] IMVAST-OCAMI Joint Conference, Osaka Metropolitan University, Japan	May 2024
[17] Geometry and Topology Seminar, KIAS, Korea	Mar 2024
[16] Workshop on Hyperbolic Geometry of Numbers, HCMC, KIAS, Korea	Dec 2023
[15] JSPS-VAST Workshop, FPT University Quy Nhon, Vietnam	Sep 2023
[14] Mathematics workshop and related fields, Ha Tinh University, Vietnam	Apr 2023
[13] JSPS-VAST Bilateral Joint Research Project Workshop, IMVAST	Dec 2022
[12] Algebra and Number Theory Seminar, IMVAST	Nov 2022
[11] Scientific conference on Geometry, Analysis and Applications, Phu Tho, Vietnam	Oct 2022
[10] Topology and Geometry Seminar, NUS, Singapore	Sep 2022
[9] Scientific workshop for young researchers, Vinh Phuc, Vietnam	Aug 2022
[8] Japan-Vietnam Singularity Seminar (Online), IMVAST	Apr 2022
[7] Mostly Teichmüller Spaces Seminar (Online), Queens College, USA	Mar 2022
[6] Ph.D. Defense, Geometry Winter Workshop, Luxembourg	Jan 2022
[5] Topology and Geometry Seminar (Online), IMVAST	Nov 2021
[4] Topology and Geometry Seminar (Online), VNU, Ho Chi Minh City, Vietnam	Oct 2021
[3] PhD Away Days (Online), University of Luxembourg, Luxembourg	Oct 2020
[2] PhD Seminar (Online), University of Luxembourg, Luxembourg	May 2020
[1] School on Geometry and Dynamics on Hyperbolic Surfaces, Switzerland	Aug 2018

Services

[3] Co-organizing Geometry and Dynamics in Low Dimensions Workshop, Hanoi Jan 2025
(with S. Kim, T. Koberda, C. Leininger, S. P. Tan, T. T. Nguyen, T. D. Do, V. T. Tran)

[2] Co-organizing Geometry, Topology & Dynamics Seminar, NUS, Singapore 2024 - 2026
(with Tengren Zhang, and Daren Wei)

[1] Co-writing a short comic story: Maths, Magic and Maryam 2020

Advising

[3] Undergrad student: Phuc Thinh Dang — 2026 - now

[2] PhD student: Van Nguyen — 2025 - now

[1] Vietnamese REU students: Trong Toan Dao, and Anh Tuan Nguyen — 2023 - 2024
(Co-supervised with Ser Peow Tan)

Teaching

[5] Modern Geometry, Master program in IMVAST — 2023

[4] Probability, Applied Math and Data Science students, Vietnam France University — 2022

[3] Linear Algebra (T.A.), University of Engineering and Technology — 2022

[2] Analysis for Applications (T.A.), University of Luxembourg — 2018 - 2020

[1] Elementary Mathematics, for kids (5 - 10 years old), Mathnasium Vietnam — 2012 - 2013

Awards

[1] Excellent Doctoral Thesis Award at the University of Luxembourg — 2022

Grants

[5] Singapore Academies South-East Asia Fellowship hosted in NUS, Singapore — 2024 - 2026

[4] NAFOSTED (2 years), Vietnam — 2024 - 2026

[3] Research Grant (2 years), IMVAST, Vietnam — 2022 - 2024

[2] PRIDE Grant (4 years), Fonds National de la Recherche, Luxembourg — 2018 - 2022

[1] Master 2 Excellence Scholarship (10 months) — 2016 - 2017
J.A. Dieudonné Mathematics Laboratory, Nice, France

References

Prof. Ara Basmajian
Hunter College, CUNY
abasmajian@gc.cuny.edu

Prof. François Labourie
EPFL-Institute of Mathematics
francois.labourie@epfl.ch

Prof. Martin Bridgeman
Boston College
bridgem@bc.edu

Prof. Hugo Parlier
University of Fribourg
hugo.parlier@unifr.ch

Prof. Sang-Hyun Kim
Korea Institute for Advanced Study
kimsh@kias.re.kr

Prof. Ser Peow Tan
National University of Singapore
mattansp@nus.edu.sg

Hobbies

Pingpong, Badminton, Hiking